At time t=0, the switch is closed to Position X.

Derive but do not solve a differential equation that could be used to determine the charge q as a function of time t on the positive plate of the capacitor with capacitance 4C. Express your answer in terms of 蔚, R, C, t, and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference information.

Derive an expression for q as a function of t by solving the differential equation that you derived in part A (i). Express your answer in terms of 蔚, R, C, t, and physical constants, as appropriate.

On the axes shown in Figure 2, **sketch** a graph of the absolute value 蝟V2R of the potential difference across the resistor with resistance 2R as a function of t, starting at t=0 until a long time tf later. The emf 蔚 of the battery is indicated on the vertical axis.

After tf, the circuit has reached steady-state conditions. The charge on the positive plate of the capacitor with capacitance 4C is Q. The switch is then closed to Position Y.

Derive an expression for the total energy U4C stored in the capacitor with capacitance 4C a long time after the switch is closed to Position Y. Express your answer in terms of C, Q, and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference information.

The circuit shown in Figure 1 consists of a battery of emf 蔚, resistors with the resistances indicated, initially uncharged parallel-plate capacitors with the capacitances indicated, and an initially open switch that can be closed to either Position X or Position Y.

The left vertical branch has a battery with emf script uppercase E.

The inner vertical branch has a capacitor with capacitance C.

The right vertical branch has a capacitor with capacitance 4 C.

The top horizontal branch has a resistor with resistance R between the left and inner vertical branches and has a resistor with resistance 2 R between the inner and right vertical branches.

The two-way open switch is at the intersection between the top horizontal branch and the inner vertical branch, between the two resistors on the top horizontal branch.

The switch is anchored to the right side of the top horizontal branch. It can close to either position X, which is on the left side of the top horizontal branch, or to position Y, which is at the top of the inner vertical branch.

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Complete the following tasks on Figure 1.

Draw an arrow at Point A that indicates the direction of the electric field due to the arc at Point A.

Draw an arrow at Point B that indicates the direction of the electric field due to the arc at Point B.

Draw an arrow at the origin O of the xy-plane that indicates the direction of the electric force that would be exerted on a small object (not shown) with negative charge $-q$ that is at the origin.

The magnitude $E0$ of the electric field at the origin of the xy-plane due to the semicircular arc is $E0=2k$ 位r.

Derive the expression provided for $E0$. Begin your derivation by writing a fundamental physics principle or an equation from the reference information.

On the axes shown in Figure 2, **sketch** a graph of the magnitude F of the electric force that would be exerted on the object with negative charge $-q$ as a function of position x along the x-axis for $x>0$. Explicitly **label** any intercepts or maxima in terms of 位, r, q, and physical constants, as appropriate.

The object with negative charge $-q$ is released from rest at the origin of the xy-plane.

Indicate whether the electric potential energy of the arc-object system increases, decreases, or remains the same after the object is released from rest.

blank_____ Increases

blank_____ Decreases

blank_____ Remains the same

Briefly **justify** your answer by referencing at least one feature of the representation from part A.

A nonconducting semicircular arc has uniform negative linear charge density -位. The arc is positioned in the xy-plane shown in Figure 1. The ends of the arc are on the y-axis at positions $y=-r$ and $y=r$.

The semicircular arc has linear charge density negative lambda and begins on the y axis at y equals r.

The semicircular arc is in the second and third quadrants. One end of the arc is at y equals r. The other end is at y equals negative r.

Point A is located at the bottom of the y axis below y equals negative r.

Point B is at the approximate coordinates (negative r over 2 comma negative r over 2).

530. In Experiment 1, students are asked to use a graph to determine whether the resistivity is constant along the length of a cylindrical circuit element that has an unknown, constant cross-sectional area. The circuit element is connected to a battery of known emf. The terminals of a voltmeter can be placed on the circuit element, as shown in Figure 1. The students also have access to a ruler.

Wires connect the battery and the ends of the circuit element, positive to the left end of the element and negative to the right.

One voltmeter lead touches the left end of the circuit element.

The other lead touches the circuit element to the right of the first lead.

Describe a procedure for collecting data that would allow the students to use a graph to determine whether the resistivity is constant along the length of the circuit element. Include any steps necessary to reduce experimental uncertainty.

Describe how the collected data could be graphed and how that graph would be analyzed to make the determination.

In Experiment 2, the students are asked to use a graph to determine the inductance L of an inductor that is connected in an LC circuit. In each trial, a charged capacitor with variable capacitance C is connected to the inductor. The angular frequency 蝇 of the oscillation of current in the circuit is determined for five different values of C. The collected data are shown in Table 1.

Table 1

C (脑10-11 F)	蝇 (脑106 s-1)
4	7.4
8	5.1
15	3.9

25	2.9
50	2.2

Indicate two quantities, either measured quantities from Table 1 or additional calculated quantities, that could be graphed to produce a straight line that could be used to determine L.

Vertical axis: blank_____ Horizontal axis: blank_____

On the grid provided, create a graph of the quantities indicated in part C (i).

Use Table 2 to record the data points or calculated quantities that you will plot.

Clearly **label** the axes, including units as appropriate.

Plot the points you recorded in Table 2.

Draw the best-fit line for the data graphed in part C (ii).

Using the best-fit line that you drew in part C (iii), **calculate** an experimental value for L.

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Indicate whether the magnitude F of the magnetic force exerted on Wire 1 is greater in Scenario X, greater in Scenario Y, or equal in both scenarios.

Justify your answer.

Derive an expression for the magnitude of the magnetic field a distance r=b from the center of Wire 2 due to the current in Wire 2. Express your answer in terms of a, b, C, and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference information.

Indicate whether the expression that you derived in part B is or is not consistent with the answer you provided in part A. Briefly **justify** your answer.

Each scenario shows two wires, one over the other, with their centers distance H apart.

Each scenario is as follows:

Scenario 1:

The top wire is labeled Wire 1 and the bottom wire is labeled Wire 2.

Wire 1 is a solid circle that carries a current I.

Wire 2 is a hollow ring with inner radius a and outer radius b, and has a current density J equals C r cubed.

Scenario Y:

The top wire is labeled Wire 1 and the bottom wire is labeled Wire 3.

Wire 1 is a solid circle that carries a current I.

Wire 3 is a hollow ring with inner radius 2 a and outer radius b, and has a current density J equals C r cubed.

Long, solid Wire 1 carries uniform current I. In scenarios X and Y, Wire 1 is parallel to a different wire that is long and hollow. In both scenarios, the centers of the two wires are held fixed a distance H apart, as shown. The hollow wires in each scenario are described as follows.

Scenario X: Wire 2 has inner radius a and outer radius b. The magnitude of the current density in Wire 2 as a function of radial distance r is described by $J=Cr3$ for $a<r<b$, where C is a positive constant.

Scenario Y: Wire 3 has inner radius $2a$ and outer radius b. The magnitude of the current density in Wire 3 as a function of r is described by $J=Cr3$ for $2a<r<b$, where C is the same constant as in Scenario X.

---

488. Which pair of diagrams best represents the direction, if any, of $\vec{F}$ and the distribution of charge on the sphere when the rod is held close to the sphere?

A negatively charged rod is held close to but does not touch one side of a neutral conducting sphere that hangs from a rigid, insulated support, as shown.

The electric force exerted on the sphere, if present, is $\vec{F}$

A.

B.

C.

D.

---

489. What is the electric potential energy of the two-sphere

Two nonconducting spheres each have a net charge that is uniformly distributed throughout the sphere. One sphere has a radius of 0.006 m and a charge of $+8\times10^{-9}$ C. The other sphere has a radius of 0.007 m and a charge of $+5\times10^{-9}$ C. The centers of the spheres are 0.4 m apart.

A. 5×10^{-7} J

B. 9×10^{-7} J

C. 1×10^{-4} J

D. 2×107 J

---

490. Which expression is equal to the time constant of Circuit 2?

488:C
489:B
490:B
491:B
492:C
493:C
494:C
495:B
496:B
497:D
498:C
499:D
500:C
501:B
502:D
503:A
504:B
505:B
506:D
507:C
508:C
509:C
510:D
511:C
512:B
513:B
514:D
515:C
516:C
517:A
518:D
519:A
520:C
521:C
522:D
523:D
524:B
525:A
526:D
527:B

In Circuit 1, a battery, a resistor of resistance R, and a capacitor of capacitance C are connected in series. The time constant of Circuit 1 is τ. In Circuit 2, a battery, a resistor of resistance R, and N capacitors, each of capacitance C, are connected in series.

A. $\tau N2$

B. τN

C. $N\tau$

D. $N2\tau$

---

491. Which of the following correctly relates I1, I2, and I3?

The circuit shown consists of a battery and three identical resistors. Currents I1, I2, and I3 are indicated.

The circuit has two horizontal branches and three vertical branches.

From top to bottom, the left vertical branch has a resistor followed by a battery.

The inner vertical branch has a resistor, and below that resistor a current I subscript 2 is directed downward.

The right vertical branch has a resistor.

Between the left and inner vertical branches, the top horizontal branch has a current I subscript 1 that is directed to the right.

Between the inner and right vertical branches, the bottom horizontal branch has a current I subscript 3 that is directed to the left.

A. I1=I2=I3

B. I1=I2+I3

C. I2=I1+I3

D. I3=I1+I2

---

492. Which of the following is equal to the current in one resistor a long time after the switch is closed?

A battery of emf ε, an inductor of inductance L, two identical resistors, and an initially open switch are connected in series to form a circuit. The time constant of the circuit is τ.

A. Zero

B. $\varepsilon L \tau$

C. $\varepsilon \tau L$

D. $2 \varepsilon \tau L$

---

493. Which of the following is equal to the electric potential at the center of Sphere Y after the spheres are moved very far

Initially isolated conducting Sphere X has radius R and charge Q. Initially isolated conducting Sphere Y has radius 3R and is neutral. The spheres are brought together until they touch. Later, the spheres are moved very far apart.

A. Zero

B. kQ6R

C. kQ4R

D. 3kQ4R

---

494. Which of the following correctly compares F2, F3, and F4?

Small spheres 1, 2, 3, and 4 with the charges indicated are held fixed along a line. The centers of adjacent spheres are a distance r apart, as shown.

The magnitudes of the net electric forces exerted on spheres 2, 3, and 4 are F2, F3, and F4, respectively.

A. F2>F3>F4

B. F2=F3>F4

C. F4>F2=F3

D. F4>F3>F2

---

495. What is the approximate induced emf in the loop?

A long current-carrying solenoid has 380 turns per meter. A circular conducting loop that has a radius of 0.01 m is in the center of the section of the solenoid shown.

The area vector of the loop is parallel to the central axis of the solenoid. The current in the solenoid increases by 0.60 A in 0.090 s at a constant rate.

A. 8.0×10^{-9} V

B. 1.0×10^{-6} V

C. 2.0×10^{-6} V

D. 2.0×10^{-4} V

---

496. What is the kinetic energy of the sphere after the sphere moves through ΔV?

A sphere with a charge of $+8.0\times10^{-9}$ C has an initial kinetic energy of 1.2×10^{-6} J. The sphere then moves through an electric potential difference of $\Delta V=-50$ V. The only force exerted on the sphere is the electric force.

A. 4.0×10^{-7} J

B. 8.0×10^{-7} J

C. 1.2×10^{-6} J

D. 1.6×10^{-6} J

---

497. Which graph could represent the electric potential V along the y-axis as a function of position y?

A uniform electric field of magnitude E is directed in the $+x$-direction, as shown.

A.

The line begins in the third quadrant at the left end of the horizontal axis and near the bottom of the vertical axis, and slants upward and to the right.

It passes through the origin, continues upward and to the right, and ends in the first quadrant at the right end of the horizontal axis and near the top of the vertical axis.

B.

The line begins in the second quadrant at the left end of the horizontal axis and near the top of the vertical axis, and slants downward and to the right.

It passes through the origin, continues downward and to the right, and ends in the fourth quadrant at the right end of the horizontal axis and near the bottom of the vertical axis.

C.

The parabola opens upward.

The parabola begins near the left end of the horizontal axis and at the top of the vertical axis, and moves downward and to the right rapidly at first and then gradually.

It reaches a minimum and becomes horizontal below the origin at the bottom of the vertical axis, then moves upward and to the right gradually at first and then rapidly.

It ends near the right end of the horizontal axis and at the top of the vertical axis.

D.

The line begins above the left end of the horizontal axis, halfway up the vertical axis from the origin.

It moves horizontally to the right, crossing the vertical axis, and continues until it ends above the right end of the horizontal axis.

---

498. Which expression is equal to the absolute value of the potential difference between labeled points X and Y?

The circuit shown consists of a battery of emf ε and four identical capacitors.

The top horizontal branch has a battery with emf script uppercase E.

From left to right, the bottom horizontal branch has a point labeled X, a capacitor, and a point labeled Y.

To the right of point Y, the bottom horizontal branch splits into two horizontal subbranches.

The top subbranch has two capacitors. The bottom subbranch has one capacitor.

A. ε 3

B. 2 ε 5

C. 3 ε 5

D. 2 ε 3

---

499. Which figure represents possible magnetic field lines and provides a valid justification?

Figures 1 and 2 both show a spherical Gaussian surface in the region of a field. The field lines are indicated in each figure.

A. Figure 1, because the field in Figure 1 could originate from a U-shaped bar magnet that is inside of the surface.

B. Figure 1, because the lines in Figure 1 get farther apart as the radial distance from the surface increases.

C. Figure 2, because the lines in Figure 2 are parallel and only magnetic fields can have parallel field lines.

D. Figure 2, because the number of field lines that enter the surface in Figure 2 is equal to the number of field lines that exit the surface.

---

500. Which of the following correctly describes the induced emf ε in the loop as t increases?

A conducting loop is held fixed in a time-varying magnetic field that is directed along the x-axis. The area vector of the loop is directed along the x-axis. The x-component of the field as a function of time t is described by $B_x = \alpha - \beta t$, where α and β are constants.

A. ε increases.

B. ε decreases.

C. ε is constant and nonzero.

D. ε is constant and zero.

---

501. What is the equivalent resistance of the circuit after the switch has been closed for a long time?

The circuit shown consists of a battery; a capacitor of capacitance *C*; resistors of resistances *R*, 2R, and 3R; and a switch that has been closed for a long time.

The circuit has three vertical branches and two horizontal branches.

The left vertical branch has a resistor with resistance R.

The inner vertical branch has a capacitor with capacitance C.

The right vertical branch has a battery.

The top horizontal branch has a resistor with resistance 2 R between the inner and right vertical branches.

From left to right, the bottom horizontal branch has a resistor with resistance 3 R and a closed switch between the inner and right vertical branches.

A. Zero

B. 11R5

C. 5R

D. 6R

---

502. The switch is then opened. The absolute value of the potential difference across the capacitor as a function of time is ΔVC. Which of the following indicates how ΔVC changes after the switch is opened, if at all, and provides a valid

The circuit shown consists of a battery; a capacitor of capacitance *C*; resistors of resistances *R*, 2R, and 3R; and a switch that has been closed for a long time.

The circuit has three vertical branches and two horizontal branches.

The left vertical branch has a resistor with resistance R.

The inner vertical branch has a capacitor with capacitance C.

The right vertical branch has a battery.

The top horizontal branch has a resistor with resistance 2 R between the inner and right vertical branches.

From left to right, the bottom horizontal branch has a resistor with resistance 3 R and a closed switch between the inner and right vertical branches.

A. ΔVC is constant because the charge on the positive capacitor plate is constant.

B. ΔVC is constant because the capacitor is in parallel with the battery.

C. ΔVC decreases because the capacitor is in series with the battery.

D. ΔVC decreases because the charge on the positive capacitor plate decreases.

---

503. A solid conducting sphere has radius *a* and net charge *Q*. Which expression is equal to the magnitude of the electric field due to the sphere at a distance 3a from the surface of the

A. 116kQa2

B. 19kQa2

C. 14kQa2

D. 13kQa2

---

504. Which expression is equal to the electric potential at Point

A thin ring of radius *a* has uniformly distributed net charge *Q*. Point P is a distance *b* above the center of the ring, as shown.

A. 12πkQa2+b2

B. kQa2+b2

C. 12πkQa

D. kQa2+b2

---

505. Which expression is equal to the energy stored in the capacitor after steady-state conditions are established in

In Circuit 1, a battery of emf ε, a resistor of resistance *R*, and an initially uncharged capacitor of capacitance *C* are connected in series. After steady-state conditions are established, the energy stored in the capacitor is *U*. In Circuit 2, a battery of emf 2ε, a resistor of resistance 3R, and an initially uncharged capacitor of capacitance *C* are connected in

A. 2U

B. 4U

C. 6U

D. 9U

---

506. Which expression is equal to the resistance of the wire?

A straight wire is placed along the *x*-axis with one end at the origin and the other end at position x=L. The wire has cross-sectional area *A*. The resistivity of the wire as a function of *x* is described by ρ=αx, where α is a constant.

A. αL2A

B. αLA

C. αL2A

D. αL22A

---

507. What is the approximate magnitude of the net electric field at Point P?

Small charged spheres S and T are held fixed on the *xy*-coordinate system shown. The position and charge of each sphere are indicated. The position of Point P is also indicated.

Sphere S is at the origin with coordinates 0.00 meters comma 0.00 meters and has a charge of positive 7 nanocoulombs.

Sphere T has coordinates 0.05 meter comma 0.03 meter and has a charge of negative 4 nanocoulombs.

A point labeled P is located at 0.05 meter comma 0.0 meter on the x axis, directly below Sphere T.

A. 2×103 N/C

B. 1×104 N/C

C. 5×104 N/C

D. 7×104 N/C

---

508. A positive test charge is held fixed at Point P. Which of the following correctly indicates the signs of the x- and y-components of the net electric force $\vec{F}$ exerted on the test charge and provides a valid justification?

Small charged spheres S and T are held fixed on the xy-coordinate system shown. The position and charge of each sphere are indicated. The position of Point P is also indicated.

Sphere S is at the origin with coordinates 0.00 meters comma 0.00 meters and has a charge of positive 7 nanocoulombs.

Sphere T has coordinates 0.05 meter comma 0.03 meter and has a charge of negative 4 nanocoulombs.

A point labeled P is located at 0.05 meter comma 0.0 meter on the x axis, directly below Sphere T.

A. $\vec{F}$ has negative x- and y-components because $\vec{F}$ is in the same direction as the net electric field at Point P.

B. $\vec{F}$ has negative x- and y-components because $\vec{F}$ is in the opposite direction of the net electric field at Point P.

C. $\vec{F}$ has positive x- and y-components because $\vec{F}$ is in the same direction as the net electric field at Point P.

D. $\vec{F}$ has positive x- and y-components because $\vec{F}$ is in the opposite direction of the net electric field at Point P.

---

509. An electron moves undeflected through a region that has perpendicular electric and magnetic fields of magnitudes 800 V/m and 4.0×10-3 T, respectively. What is the speed of the electron?

A. 5.0×10-6 m/s

B. 3.2 m/s

C. 2.0×105 m/s

D. 1.7×107 m/s

---

510. What is the magnitude of the magnetic force exerted on the right side of the loop in Scenario 2?

In Scenario 1, a rigid wire loop with nonnegligible resistance is moving to the right at constant speed v while entering an external uniform magnetic field that is out of the page, as shown in Figure 1. In Scenario 2, the loop is moving to the right at constant speed 2v while entering the field, as shown in Figure

In each figure, a rectangular loop moves horizontally right towards a region of a uniform magnetic field, with the loop partially in the magnetic field and the field directed out of the page.

In Scenario 1, the loop moves with velocity v.

In Scenario 2, the loop moves with velocity 2 v.

In Scenario 1, the magnitude of the magnetic force exerted on the right side of the loop is F.

A. Zero

B. F2

C. F

D. 2F

---

511. What is the approximate capacitance of the capacitor with the material inserted between the plates?

An air-filled, charged parallel-plate capacitor is isolated. The area of each plate is 2.0×10-2 m2. The plates are separated by a distance of 4.0×10-4 m. The magnitude of the electric field between the plates is 5.0×10-5 V/m. Later, a dielectric material is fully inserted between the plates and the new magnitude of the electric field between the plates is 3.0×10-

A. 2.7×10-10 F

B. 4.4×10-10 F

C. 7.4×10-10 F

D. 8.8×10-10 F

---

512. What is the approximate value of the ratio Φ2Φ1?

Uniform electric field E→ is directed along the horizontal and through a loop that has area vector A→ that is initially directed 37° above the horizontal, as shown in Figure 1. The loop is then rotated so that A→ is directed 53° above the horizontal, as shown in Figure 2. The electric fluxes through the loop in figures 1 and 2 are Φ1 and Φ2, respectively.

Each figure shows a side view of a loop, represented by a line segment, that has an area vector A perpendicular to that line segment.

Uniform electric field vector E is directed horizontally right, through the loop.

In Figure 1, the loop is oriented so that its area vector A points upward and to the right at a 37 degree angle above

In Figure 2, the loop is rotated counterclockwise so that its area vector A points upward and to the right at a 53 degree angle above horizontal.

A. 0.60

B. 0.75

C. 0.80

D. 1.3

---

513. Which of the following indicates how the magnitude *a* of the acceleration of the electron changes as the electron moves and provides a valid justification?

Points A and B are on the equipotential lines shown. An electron moves freely from Point A to Point B.

From left to right, the lines are labeled negative 10 volts, negative 5 volts, 0 volts, and 5 volts.

The distance between the 0 volt and 5 volt lines is the

The distance between the negative 5 volt and 0 volt lines is about one half the of the distance between the 0 volt and 5 volt lines.

The distance between the negative 10 volt and negative 5 volt lines is about one half the of the distance between the negative 5 volt and 0 volt lines.

A point labeled A is shown on the negative 10 volt line.

A point labeled B is shown on the 5 volt line.

A. *a* increases because the electron moves from a negative to a positive electric potential.

B. *a* increases because the change in the electric potential per unit distance increases as the electron moves.

C. *a* decreases because the magnitude of the electric field decreases as the electron moves.

D. *a* decreases because the total change in the electric potential from Point A to Point B is positive.

---

514. What is the ratio L2L1 ?

Long, air-filled solenoids 1 and 2 have the same diameter and length. Solenoid 1 has 100 loops. Solenoid 2 has 200 loops. The inductances of solenoids 1 and 2 are L1 and L2, respectively.

A. 14

B. 12

C. 2

D. 4

---

515. Which of the following correctly compares F1, F2,

A wire carrying current I is in the xy-plane. The current is directed 30° above the x-axis, as shown. The wire is in an external uniform magnetic field of magnitude B in scenarios 1, 2, and 3, as described.

Scenario 1: The field is directed in the +x - direction.

Scenario 2: The field is directed in the +y - direction.

Scenario 3: The field is directed in the +z - direction.

A coordinate basis indicates that the positive x direction points horizontally to the right, the positive y direction points vertically upward, and the positive z direction points out of the page.

The wire passes diagonally through the origin at an angle of 30 degrees above horizontal.

A current I is directed upward and to the right through the

The magnitudes of the magnetic forces exerted on the wire in scenarios 1, 2, and 3 are F1, F2, and F3, respectively.

A. F1>F2>F3

B. F1>F3>F2

C. F3>F1>F2

D. F3>F2>F1

---

516. Which expression is equal to the absolute value of the electric flux through the shaded top half of the Gaussian

A small charged sphere is at the center of a spherical Gaussian surface of radius r. The top half of the Gaussian surface is shaded, as shown.

The magnitude of the electric field at the Gaussian surface is

A. 12πr2E

B. πr2E

C. 2πr2E

D. 4πr2E

---

517. Which of the following correctly compares IX, IY, and IZ?

Circuits X, Y, and Z each consist of a battery of emf ε and three resistors. The resistances of the resistors and the currents in the right branches of circuits X, Y and Z are

In each circuit, there are two horizontal branches and three vertical branches.

Each circuit consists of one battery and three resistors, with each left vertical branch having a battery with emf script uppercase E.

Each diagram is as follows:

Circuit X:

The inner vertical branch has a resistor with resistance

From top to bottom, the right vertical branch has a resistor with resistance 3 R, followed by a current I subscript X directed downward, followed by a resistor with resistance 2 R.

Circuit Y:

The inner vertical branch has a resistor with resistance 3 R.

From top to bottom, the right vertical branch has a resistor with resistance R, followed by a current I subscript y directed downward, followed by a resistor with resistance 2 R.

Circuit Z:

The inner vertical branch has a resistor with resistance 2 R.

From top to bottom, the right vertical branch has a resistor with resistance R, followed by a current I subscript Z directed downward, followed by a resistor with resistance 3 R.

A. IX>IZ>IY

B. IX>IY>IZ

C. IY>IZ>IX

D. IZ>IX>IY

---

518. Which of the following correctly compares VR, VS,

Three small nonconducting spheres with the charges indicated are evenly spaced and held fixed on the x-axis. Points R, S, and T are evenly spaced such that points S and T are midway between the centers of the spheres, as shown.

From left to right, the objects along the x axis are as

A point labeled R

A sphere with charge positive Q

A point labeled S

A sphere with charge positive Q

A point labeled T

A sphere with charge negative Q

The electric potentials due to the three spheres at points R, S, and T are VR, VS, and VT, respectively.

A. VR>VS>VT

B. VS>VR>VT

C. VT>VR>VS

D. VT>VS>VR

---

519. Which expression is equal to the magnitude of the magnetic force exerted on the bar as a function of t as the bar moves?

A bar with resistance R is moved in the +x - direction by an external force along a U-shaped loop of conducting rails. The rails are a distance D apart. The bar and the rails are in a uniform magnetic field of magnitude B that is directed out of the page, as shown.

The position of the bar as a function of time t is described by x=α t2, where α is a positive constant.

A. 2αB2D2tR

B. αB2D2t2R

C. 2αB2D2t

D. αB2D2t2

---

520. What is the approximate radius of Segment Y?

A wire is made of cylindrical segments X and Y that are connected in series. The uniform current density in Segment X is 3.8×106 A/m2, and the uniform current density in Segment Y is 1.9×106 A/m2. The radius of Segment X is 5.0×10-4 m.

A. 5.0×10-7 m

B. 3.5×10-4 m

C. 7.1×10-4 m

D. 1.0×10-3 m

---

521. What is the absolute value of the electric flux through the Gaussian sphere?

An object with a charge of +3.54×10-5 C is enclosed by a Gaussian sphere that has a radius of 1.0 m. A second object, which has a charge of -3.54×10-5 C, is 2.0 m from the center of the same Gaussian sphere.

A. Zero

B. 3.2×105 N•m2C

C. 4.0×106 N•m2C

D. 8.0×106 N•m2C

---

522. A wire of radius *a* carries current *I* with uniform current density. Which expression is equal to the magnitude of the magnetic field at a distance *r* from the center of the wire,

A. μ0I4πr

B. μ0I2πr

C. μ0Ir4πa2

D. μ0Ir2πa2

---

523. Which of the following provides a diagram that represents the correct placement of the ammeter in the circuit and correctly compares *I* to εR?

A circuit consists of a battery of emf ε, a resistor of resistance *R*, and a nonideal ammeter that has a resistance that is less than *R*. A student is asked to connect the ammeter in the circuit to measure the current *I* in the resistor.

A.
Circuit **Comparison of *I* to εR**
I=εR
B.
Circuit **Comparison of *I* to εR**
I<εR
C.
Circuit **Comparison of *I* to εR**
I=εR
D.
Circuit **Comparison of *I* to εR**

I<εR

---

524. Which expression is equal to the magnitude of E→?

At time t=0, a small sphere with mass *M* and positive charge +Q is moving with speed *v* to the right midway between two large, charged parallel plates. The plates are a distance *D*

The electric field E→ between the plates is uniform and constant. Gravitational effects are negligible. The sphere reaches one plate at time tf.

A. 12MDQtf2

B. MDQtf2

C. 12Mv2QD

D. Mv2QD

525. Which of the following correctly compares $\Phi 2$ to $\Phi 1$?

In Scenario 1, a circular wire loop of radius r is placed in a uniform magnetic field of magnitude B. In Scenario 2, a circular wire loop of radius 2r is placed in a uniform magnetic field of magnitude B2. In both scenarios, the area vectors of both loops are in the same direction as the magnetic fields. The magnetic fluxes through the loops in scenarios 1 and 2 are $\Phi 1$ and $\Phi 2$, respectively.

A. $\Phi 2>\Phi 1$

B. $\Phi 2<\Phi 1$

C. $\Phi 2=\Phi 1>0$

D. $\Phi 2=\Phi 1=0$

---

526. Which expression is equal to the experimental value of mass

Small spheres, each of the same unknown mass m but with different amounts of charge q, are launched at speed v toward a region with an external uniform magnetic field of magnitude B. The field is perpendicular to the direction that the spheres are launched. The spheres move along circular paths after entering the field, as shown in Figure 1. Figure 2 shows values of the radius r of each circular path for each tested value of 1/q. The slope of the dashed best-fit line in Figure 2 is s.

Each figure is as follows:

Figure 1:

A sphere of mass m and charge q moves horizontally to the right with velocity v, entering the top left of a uniform magnetic field B.

The field is directed out of the page, and the left side of the field is a vertical line.

The sphere enters the magnetic field, then moves downward and to the right along a semicircular path with radius r through the interior of the field, and would exit the field moving horizontally to the left.

Figure 2:

A scatterplot with a line of best fit in the first quadrant of a coordinate plane.

The horizontal axis is labeled 1 over q, times 10 to the power of 18 inverse coulombs, and the numbers 0 through 7, in increments of 1, are indicated.

The vertical axis is labeled r in meters, and the numbers 0.000 through 0.005, in increments of 0.001, are

The line of best fit begins near the origin, slants upward and to the right, and ends at approximately the point with coordinates 6.2 comma 0.005.

A. s

B. 1s

C. Bsv

D. sBv

---

527. Which diagram best represents the modified circuit?

The circuit shown consists of a battery of emf ε, a resistor of resistance R, and a parallel-plate capacitor of capacitance C.

Charge Q is stored on the positive capacitor plate when steady-state conditions are established. The circuit is then modified and reaches steady state with charge 2Q stored on the positive

A.

B.

C.

D.

---

528 – 531 物理题详解（完整版）

下面按题号逐小题书写中文详解。为方便阅读，公式用 \(\LaTeX \) 行内排版，必要处给出推导的核心步骤；对需作图的小题，用文字描述应呈现的曲线或箭头方向即可——考卷上只要按此描述落笔即可得

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528 RC 电路与电容能量

电路框架回顾

电池电动势 \(\varepsilon \) 在最上方水平支路上依次经过电阻 \(R\) 与 \(2R\)。
内侧竖直支路是电容 \(C\)，右侧竖直支路是电容 \(4C\)。
开关位于 \(R\) 与 \(2R\) 之间的节点 *P*，可拨到 **X**（与左侧导线相接）或 **Y**（与竖直电容 *C* 的顶端相接）。

A(i) 写出 \(4C\) 电容板上电荷 \(q(t)\) 的微分方程（不必求解）

闭合到 **X** 时，回路列式

\[\varepsilon - iR - i(2R) - \frac{q}{4C}=0 \]

又 \(i = \dfrac{\mathrm dq}{\mathrm dt} \)。整理得

3R dq/dt + q/(4C) = ε

A(ii) 求 \(q(t)\)

写成

\[\frac{\mathrm dq}{\mathrm

设时间常数 \(\tau = 12RC \)，得到

q(t)=4Cε[1-e^{-t/(12RC)}]

A(iii) \(|ΔV_{2R}|\) 随 *t* 变化曲线

初值 \(|ΔV_{2R}(0)| = \tfrac{2ε}{3}\)
指数衰减：\(|ΔV_{2R}(t)| = \tfrac{2ε}{3}e^{-t/(12RC)}\)

图像：自 $t=0$ 处取值 $2\varepsilon/3$，单调向 0 渐近的指数曲线。

B 切换到 Y 后很久的能量 U_{4C}

切换瞬间：
C 原本未充电，$4C$ 带正电 $Q (=4C\varepsilon)$。

电流经 $2R$ 短暂流动后，两电容顶板电势相等：$q_C/C = q_{4C}/(4C) \Rightarrow q_{4C}=4q_C$。
总电荷守恒：$q_C+q_{4C}=Q \Rightarrow 5q_C=Q$。
$q_C = Q/5, \; q_{4C}=4Q/5$
能量： $U_{4C}=\dfrac{q_{4C}^2}{2\cdot 4C}=\dfrac{2Q^2}{25C}$。

529 带电半圆弧

A 箭头方向（文字说明）

	位置	电场方向	理由
A	($y<-r$)	向 上	径向分量沿 $+y$
B	($-r/2,-r/2$)	约 右下	水平向原点，垂直向
O		电场向左 负荷受力向右	弧在左，负电荷受反向力

B 推导 $E_O = 2k\lambda/r$

$\mathrm dq = \lambda r\,\mathrm d\theta$
$\mathrm dE = k\lambda\,\mathrm d\theta / r$
水平方向：$E_O = \int_{-\pi/2}^{\pi/2} k\lambda\cos\theta/r\,\mathrm d\theta = 2k\lambda/r$

C $|F|-x$ 图像

$x=0$ 处峰值 $|q|\cdot 2k\lambda/r$
随 x 增大单调下降，渐近于 0。

D 势能变化

下降。负电荷被推向 +x，电势增大而 $q<0 \Rightarrow U=qV$ 减小。

530 实验设计与数据处理

实验 1 ：检验电阻率是否恒定

(A) 数据采集

用尺子在导体表面每隔固定距离标记 x_i。
一支电压表探针固定于左端，另一支依次接触各标记，记录 ΔV_i
每点交换表笔复测取平均，减少接触电阻误差。
保持电池电压恒定，必要时停顿冷却以防温升。

(B) 绘图分析

作图 ΔV 对 x：若呈直线且过原点，则电阻率 ρ 恒定。斜率给出 $\rho I/A$。

实验 2：测电感 L

LC 振荡公式 $\omega^2 = 1/(LC)$。

(C i) 可线性化坐标

纵轴 $Y = 1/\omega^2$，横轴 $X = C$。

(C ii–iii) 作图要点

计算 $1/\omega^2$ 后绘制 (Y vs X) 五点。
标明单位并绘制最佳拟合线。

(D) 由斜率求 L

拟合斜率即 L，示例计算约 0.41 mH。

531 并行导线与安培力

A 受力大小比较

下方空心导线电流

$$I_{\text{tot}} = 2\pi C \int_{r_0}^{b} r^3\,dr = \frac{2\pi C}{5}\,(b^5 -$$

情景 X 内半径 a，情景 Y 内半径 2a，故 $I_X > I_Y$ ⇒

B 距中心 b 处磁场

安培环路：$B\cdot 2\pi b = \mu_0 I_{\text{enc}}$，得

$B(b) = \mu_0 C(b^5 - a^5)/(5b)$

C 一致性说明

若 a→2a (情景 Y)，则 B、F 均减小，与 (A) 一致。

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